

GEOG 4/5/7 9073: Environmental Analysis in R

Week 05.01: R as a GIS

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Today's schedule

- Open discussion
- Some basic spatial operations
- Activity

Anything to discuss? Questions?

This week's focus: simple geospatial operations and getting comfortable with them

today's setup

- there's new data in the `./data/` directory of the repository, and we'll be using some new packages as well

```
library(tidyverse)
library(sf)
library(GISTools)
library(tmap)
```

Let's start with some simple data

Break it down first

```
# let's start with some data

streams <- sf::read_sf("./data/Streams_303_d_.shp")
tm_shape(streams) + tm_lines()
```

what happened?

More data...

```
counties <- sf::read_sf("./data/County_Boundaries-Census.shp")  
counties_areas <- sf::st_area(counties)
```

and what happened here?

If you wanted to, how would you add the areas back to the sf data.frame?

**Let's review... how would you use your data and functions to "get"
ONLY Lancaster County?**

Something like this:

which function are we using, and how does it work?

```
lc <- counties %>% dplyr::filter(., NAME10 == "Lancaster")
```

How would we find only those 303d streams in Lancaster County?

Something like this:

```
lc_303ds <- sf::st_intersection(streams, lc)
```

Plot them:

Here's some options. Let's break them down - how are they different?

```
tm_shape(lc_303ds) + tm_lines()  
tm_shape(lc_303ds) + tm_lines(col = "blue")  
  
tm_shape(lc_303ds) + tm_lines(col = "Waterbody_")
```

What happened?

Buffers: a simple spatial operation

- What's a buffer?

Let's try one

Break down the code...

```
buffs <- sf::st_buffer(lc_303ds, dist = 1000)
tm_shape(buffs) + tm_polygons(col = "Waterbody_")
```

What does it look like?

How MUCH did we buffer the streams?

Let's try some point data

```
# first read the state parks shapefile
parks <- sf::read_sf("../data/State_Park_Locations.shp")

# subset to lancaster county
lc_parks <- sf::st_intersection(parks, lc)

# plot them
tm_shape(lc_parks) + tm_dots(col = "AreaName", size = 1)
```

Plotting multiple layers using `tm`

```
tm_shape(lc_303ds) + tm_lines(col = "Waterbody_") +  
  tm_shape(lc_parks) + tm_dots(col = "AreaName", size = 1)
```

What happened?

Your final task

- find all of the state parks within 0.5 miles of a 303d stream
- plot just those points WITH the corresponding stream segment
- use color to distinguish the points and stream segments

GO!

For this week

- Chapters 5 & 6
- Practice, practice, practice
- Lab 01 due